

1.a $n, m \in \mathbb{N}$ with $1 \leq m \leq n$; $K = (k_1, \dots, k_m) \in \mathbb{R}^m$ be a non-zero 1D filter; convolution map $T_K: \mathbb{R}^n \rightarrow \mathbb{R}^{n-m+1}$ by (no padding)

$$(T_K(x))_t = \sum_{i=1}^m k_i x_{t+1-i}, \quad t=1 \dots n-m+1$$

$A_K \in \mathbb{R}^{(n-m+1) \times n}$ denote the matrix such that $T_K(x) = A_K x$

1.a) write A_K explicitly, prove A_K is Toeplitz.

The t th row of A_K is nothing but the output of $(T_K(x))_t$

$(k_1, k_2, \dots, k_m)$ will be the coefficients of $(x_t, x_{t+1}, \dots, x_{t+m-1})$ respectively in the t th row.

A_K has $n-m+1$ rows and n columns

$$A_K = \begin{matrix} & \begin{matrix} \text{col: } 1 & 2 & \dots & m & \dots & t & \dots & n \end{matrix} \\ \begin{matrix} \text{row} \\ 0 \\ 1 \\ 2 \\ \vdots \\ t \\ \vdots \\ n-m+1 \end{matrix} & \begin{bmatrix} k_1 & k_2 & \dots & k_m & 0 & 0 & 0 & \dots & 0 \\ 0 & k_1 & \dots & k_m & 0 & 0 & \dots & 0 \\ 0 & 0 & k_1 & \dots & k_m & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 & k_1 & \dots & k_m & 0 \end{bmatrix} \end{matrix}$$

for $n=5$; $m=3$

$$A_K = \begin{bmatrix} k_1 & k_2 & k_3 & 0 & 0 \\ 0 & k_1 & k_2 & k_3 & 0 \\ 0 & 0 & k_1 & k_2 & k_3 \end{bmatrix}$$

Any matrix A_1 is toeplitz if each of its descending diagonals² from left to right contains same elements.

$$\text{example } A_1 = \begin{bmatrix} a_0 & a_1 & a_2 & a_3 & \dots \\ \dots & a_0 & a_1 & a_2 & \dots \\ \dots & \dots & \dots & a_0 & a_1 & a_2 & \dots \end{bmatrix}$$

$$\text{ie } A_1(i, j) = A_1(i', j') \text{ where } i - j = i' - j'.$$

Now in our original Matrix A_K , consider two elements at index (t, i) and (t', i') such that $t - i = t' - i'$.

for (t, i) :

* if $t \leq i \leq t + m - 1$, $A_K(t, i) = K(i - t + 1)$
ie as i takes values from t to $t + m - 1$, their respective coefficients will be $K_1, \dots, K_m$.

* otherwise $A_K(t, i) = 0$.

Similarly for (t', i')

* If $t' \leq i' \leq t' + m - 1$, then $A_K(t', i') = K(i' - t' + 1)$

* otherwise $A_K(t', i') = 0$

for $A_K(t, i)$, $A_K(t', i')$ non-zero case :-

$$\text{we know } i - t = i' - t'$$

$$i - t + 1 = i' - t' + 1$$

$$\text{ie } K_{i-t+1} = K_{i'-t'+1}$$

$$A_K(t, i) = A_K(t', i') \text{ when } t - i = t' - i'$$

for $A_K(t, i)$, $A_K(t', i')$ Zero case :-

$$A_K(t, i) = A_K(t', i') = 0 \text{ when } t - i = t' - i'$$

Hence for all (t, i) and (t', i') $A_K(t, i) = A_K(t', i')$, hence A_K is Toeplitz

1.6 if $K \neq 0$, prove A_K has full row rank, i.e. $\text{rank}(A_K) = n - m + 1$

* For any matrix, to have a full row rank, all of its rows should be linearly independent.

* As $K \in \mathbb{R}^m$, in order for $K \neq 0$, at least one of the m dimensions of K should have a non-zero value.

* Let j be the 1^{st} index where $K_j \neq 0$ in K . i.e. all the value of K_i where $i < j$ are zero.

(where $j \in (1 \dots m)$)

* we know, for row t , all the values of $A_K(t, i)$ will have zero values where $i = 1 \dots t-1$. In other words,

K_1 is the coefficient of column t for row t

K_2 is the coefficient of column $t+1$ for row t

$\dots$
 K_m is the coefficient of column $t+m-1$ for row t

* Hence the left most non-zero element is at index $t+j-1$ for row t , which is pivot position for row t
(as some of dimensions of K can be zero, but not all)
(first non-zero dimension is K_j)

* As t takes different values from 1 to $n - m + 1$,
so does the first non-zero element's index in that row t . Therefore the position of pivot ($t+j-1$) is strictly increasing as t increases from 1 to $n - m + 1$,
which means it's not possible to express any row of A_K as linear combination of others. Hence all the rows of A_K are linearly independent. Hence A_K has full rank
 $\text{rank}(A_K) = n - m + 1$

2.a

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To prove strict Concave nature of Gini Loss $G(P)$, we find the double derivate:-

$$G(P) = 2P(1-P) = 2P - 2P^2$$

$$G'(P) = 2 - 4P \text{ (1st derivative)}$$

$$G''(P) = -4 \text{ (2nd derivative)}$$

Since $G''(P) < 0$; G is Strictly Concave.

$$\forall P_L \neq P_R \quad \forall t \in (0,1)$$

To prove weighted Gini loss of the children cannot exceed that of Parent.

P_{par} = proportion of positive examples in parent region R_{par}

P_L = proportion of positive examples in Left Child region R_L

P_R = proportion of positive examples in Right Child region R_R

$|R_L|$ = No. of total ^{examples} that belong to Left Child region R_L

$|R_R|$ = No. of total examples that belong to Right Child region R_R

$|R_{par}|$ = No. of total examples that belong to parent region R_{par}

As Total # of positive examples is conserved

$$|R_{par}| \cdot P_{par} = |R_L| \cdot P_L + |R_R| \cdot P_R$$

$$P_{par} = \frac{|R_L| \cdot P_L}{|R_{par}|} + \frac{|R_R| \cdot P_R}{|R_{par}|}$$

for simplicity, let's call $\frac{|R_L|}{|R_{par}|}$ as C_L , $\frac{|R_R|}{|R_{par}|}$ as C_R .

$$P_{par} = C_L \cdot P_L + C_R \cdot P_R \quad \text{--- (1)}$$

Since no degenerative cases $C_L + C_R = 1$; $C_L, C_R > 0$

2. a From Jensen's inequality for a concave function f (5)

$$\sum_{i=1}^n \lambda_i f(x_i) \leq f\left(\sum_{i=1}^n \lambda_i x_i\right) \text{ where } \lambda_i \geq 0 \text{ and } \sum \lambda_i = 1.$$

Since $G(P)$ is strictly concave, $c_L + c_R = 1$, $c_L, c_R > 0$.

$$G(P_{\text{par}}) \geq c_L \cdot G(P_L) + c_R \cdot G(P_R)$$

Re substituting $c_L = \frac{|R_L|}{|R_{\text{par}}|}$ $c_R = \frac{|R_R|}{|R_{\text{par}}|}$

$$G(P_{\text{par}}) \geq \frac{|R_L|}{|R_{\text{par}}|} \cdot G(P_L) + \frac{|R_R|}{|R_{\text{par}}|} \cdot G(P_R)$$

$$|R_{\text{par}}| \cdot G(P_{\text{par}}) \geq |R_L| \cdot G(P_L) + |R_R| \cdot G(P_R)$$

i.e. the weighted Gini loss of children is strictly less than or equal to that of parent. In other words, the weighted

Gini loss of children can not exceed that of the parent.

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Q. b

i) List out the conditions on the cardinality and proportion of positive and negative samples on the children where Gini loss will stay same after the split.

If the split is made such that;

a) proportion of positive examples in left child (P_L) equals to proportion of positive examples in right child (P_R)

b) And $P_L = P_R = P_{\text{par}}$ i.e. proportion of positive examples in the parent

i.e. split doesn't change any proportions

$$P_{\text{par}} = P_L = P_R.$$

$$P_{\text{par}} = (C_L + C_R) \cdot P_L = (C_L + C_R) \cdot P_R$$

Since $(C_L + C_R) = 1$ and applying G function both sides

$$G(P_{\text{par}}) = G(P_L) = G(P_R).$$

$$G(P_{\text{par}}) = C_L \cdot G(P_L) + C_R \cdot G(P_R)$$

These two conditions listed as (a) and (b) above are the necessary conditions for the gini loss to be the same before and after the split.

Q.5

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ii) Show why these do not violate the strict concavity of the Gini loss.

The definition of strict Concavity is

$$\forall p_L \neq p_R, \forall t \in (0, 1): G(t \cdot p_L + (1-t) \cdot p_R) > t G(p_L) + (1-t) G(p_R)$$

Since $p_L = p_R$ Condition is explicitly excluded in the strict Concavity definition of Gini loss, the conditions listed in Q.5.i (that achieves same Gini loss before and after split) doesn't Violate the strict Concavity of the Gini loss.

iii) why these cases do not prevent a fully grown tree from achieving Zero Gini loss.

Note:- We are assuming there are no degenerate cases where positive and negative datapoints overlap in feature space.

By these cases, we are primarily referring to a Condition where Gini loss is same before and after the split as the split keeps the class proportions p_L, p_R, p_{par} the same.

This means the feature f that we chose and/or the threshold t we selected aren't the right ones. But by definition, the decision tree algorithm considers all possible features and all possible thresholds greedily and picks the right ones in each step until all the leaf nodes reach a stage of having a pure Gini loss i.e Gini loss of a full tree will reach zero. (0).

Miss Classification Loss $M(R_m) = M(P_m) = 1 - \max(P_m, 1 - P_m)$

In Simple term $M(p) = 1 - \max(p, 1 - p)$

As $p \in [0, 1]$, $\max(p, 1 - p) = \begin{cases} p, & \text{if } p \leq 0.5 \\ 1 - p, & \text{otherwise} \end{cases}$

So $\max(p, 1 - p)$ is a V shape piece wise linear function, which is convex. Hence $1 - \max(p, 1 - p)$ is a Concave function.

$N_m = |R_m|$, N_m^+ = no. of positive Examples in R_m

$P_m = \frac{N_m^+}{N_m}$ (ie proportion of positive Examples in R_m)

The misclassification loss of R_m

$$M(R_m) = 1 - \max\left(\frac{N_m^+}{N_m}, \frac{N_m - N_m^+}{N_m}\right)$$

$$M(R_m) = \frac{N_m - \max(N_m^+, N_m - N_m^+)}{N_m}$$

$$M(R_m) = \frac{\min(N_m - N_m^+, N_m^+)}{N_m}$$

$$M(R_{\text{par}}) = \min(N_{\text{par}}^+, N_{\text{par}} - N_{\text{par}}^+)$$

$$M(R_L) = \min(N_L^+, N_L - N_L^+)$$

$$M(R_r) = \min(N_r^+, N_r - N_r^+)$$

Case 1

⑦

If p_{par} , p_L , p_r all three are greater than 0.5, i.e. these classes have proportion of positive examples > 0.5

then

$$M(R_{par}) = N_{par} - N_{par}^+ \quad (\text{as } N_{par} - N_{par}^+ < N_{par}^+)$$

$$M(R_L) = N_L - N_L^+ \quad (\text{as } N_L - N_L^+ < N_L^+)$$

$$M(R_r) = N_r - N_r^+ \quad (\text{as } N_r - N_r^+ < N_r^+)$$

$$M(R_L) + M(R_r) = N_L - N_L^+ + N_r - N_r^+$$

$$M(R_L) + M(R_r) = (N_L + N_r) - (N_L^+ + N_r^+)$$

$$M(R_L) + M(R_r) = N_{par} - N_{par}^+$$

$$\boxed{M(R_L) + M(R_r) = M(R_{par})}$$

Case 2:-

If p_{par} , p_L , p_r all have values less than 0.5

$$M(R_{par}) = N_{par}^+; \quad M(R_L) = N_L^+, \quad M(R_r) = N_r^+$$

$$M(R_L) + M(R_r) = N_L^+ + N_r^+ = N_{par}^+ = M(R_{par})$$

$$\boxed{M(R_L) + M(R_r) = M(R_{par})}$$

Case 3:- If p_{par} (or) p_L (or) p_r have values differently than

Case 1 and Case 2, i.e. at least one of them have < 0.5 value and rest of them have > 0.5 (or) vice-versa

the misclassification error pre-split won't be same as

post-split. So having all the 3 values > 0.5 (or) having all 3 values < 0.5 is necessary to maintain same misclassification

2.6 loss pre-split and the post-split.

~~less for split and the less for - split~~

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2.d

Given $\text{Var}(T_i(x)) = \sigma^2 \forall i$, $\text{Cov}(T_i(x), T_j(x)) = \rho \forall i \neq j$

$$\text{Show } \text{Var}\left(\frac{1}{B} \sum_{i=1}^B T_i(x)\right) = \rho \sigma^2 + \frac{1-\rho}{B} \sigma^2$$

$$\text{Var}\left(\frac{1}{B} \sum_{i=1}^B T_i(x)\right) = \frac{1}{B^2} \text{Var}\left(\sum_{i=1}^B T_i(x)\right)$$

(as $\text{Var}(aX) = a^2 \text{Var}(X)$)

using alternative formula for variance

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2$$

we can write

$$\frac{1}{B^2} \text{Var}\left(\sum_{i=1}^B T_i(x)\right) = \frac{1}{B^2} \left[\underbrace{E\left[\left(\sum_{i=1}^B T_i(x)\right)^2\right]}_{\textcircled{1}} - \left(\underbrace{E\left[\sum_{i=1}^B T_i(x)\right]}_{\textcircled{2}}\right)^2 \right]$$

Let's expand $\textcircled{1}$ $E\left[\left(\sum_{i=1}^B T_i(x)\right)^2\right]$

$$= E\left[\sum_{i=1}^B T_i(x) \cdot \sum_{j=1}^B T_j(x)\right]$$

$$= E\left[\sum_{i=1}^B T_i(x)^2 + \sum_{\substack{j=1 \\ i=1 \\ i \neq j}}^B T_i(x) \cdot T_j(x)\right]$$

$$= E\left[\sum_{i=1}^B T_i(x)^2\right] + E\left[\sum_{\substack{i,j=1 \\ i \neq j}}^B T_i(x) \cdot T_j(x)\right]$$

$$E \left[\left(\sum_{i=1}^B T_i(x) \right)^2 \right] = \sum_{i=1}^B E \left[T_i(x)^2 \right] + \sum_{\substack{i,j=1 \\ i \neq j}}^B E \left[T_i(x) \cdot T_j(x) \right] \quad \text{--- (3)}^{(10)}$$

Let's expand (2) $\left(E \left[\sum_{i=1}^B T_i(x) \right] \right)^2$

$$= \left(\sum_{i=1}^B E \left[T_i(x) \right] \right)^2$$

$$= \sum_{i=1}^B E \left[T_i(x) \right] \cdot \sum_{j=1}^B E \left[T_j(x) \right]$$

$$= \sum_{i=1}^B \sum_{j=1}^B E \left[T_i(x) \right] E \left[T_j(x) \right]$$

$$\left(E \left[\sum_{i=1}^B T_i(x) \right] \right)^2 = \sum_{i=1}^B E \left[T_i(x) \right]^2 + \sum_{\substack{i,j=1 \\ i \neq j}}^B E \left[T_i(x) \right] E \left[T_j(x) \right] \quad \text{--- (4)}$$

Substituting (3) & (4)

$$\frac{1}{B^2} \text{Var} \left(\sum_{i=1}^B T_i(x) \right) = \frac{1}{B^2} \left[\sum_{i=1}^B E \left[T_i(x)^2 \right] + \sum_{\substack{i,j=1 \\ i \neq j}}^B E \left[T_i(x) T_j(x) \right] \right]$$

$$- \left[\sum_{i=1}^B E \left[T_i(x) \right]^2 + \sum_{i=1}^B E \left[T_i(x) \right] \cdot E \left[T_j(x) \right] \right]$$

$$= \frac{1}{B^2} \underbrace{\sum_{i=1}^B \left[E \left[T_i(x)^2 \right] - E \left[T_i(x) \right]^2 \right]}_{\text{Var} \left(\sum_{i=1}^B T_i(x) \right)} + \frac{1}{B^2} \underbrace{\sum_{\substack{i,j=1 \\ i \neq j}}^B \left(E \left[T_i(x) T_j(x) \right] - E \left[T_i(x) \right] \cdot E \left[T_j(x) \right] \right)}_{\text{Cov} \left(T_i(x), T_j(x) \right)}$$

2.1.8 Therefore

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$$\frac{1}{B^2} \text{Var} \left(\sum_{i=1}^B T_i(n) \right) = \frac{1}{B^2} \left[\sum_{i=1}^B \text{Var}(T_i(n)) + \sum_{\substack{i,j=1 \\ i \neq j}}^B \text{Cov}(T_i(n), T_j(n)) \right]$$

Given $\rho = \text{Corr}(T_i(n), T_j(n)) \ i \neq j$

$$\rho = \frac{\text{Cov}(T_i(n), T_j(n))}{\sqrt{\text{Var}(T_i(n))} \sqrt{\text{Var}(T_j(n))}}$$

$$\therefore \boxed{\text{Cov}(T_i(n), T_j(n)) = \rho \sigma^2} \quad - (5)$$

The # of (i, j) pairs where $i = j$ will be B .

The # of (i, j) pair where $i \neq j$ will be $B(B-1)$
($B^2 - B + B = B^2$ (total))

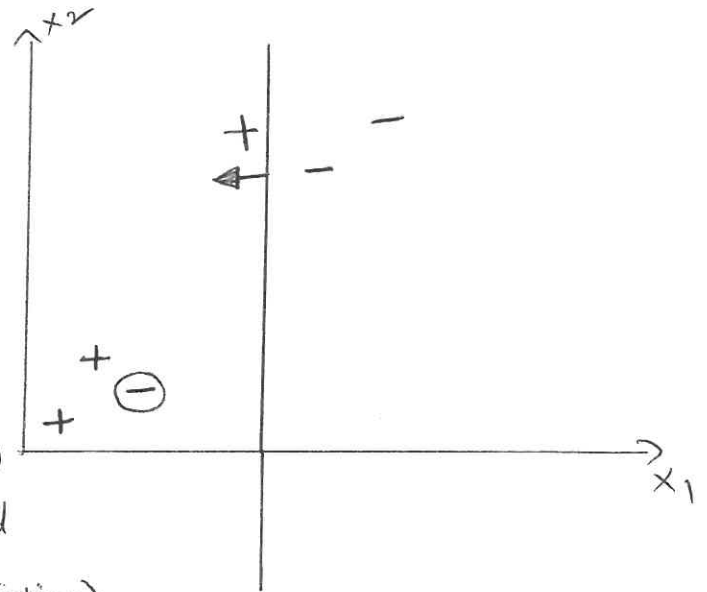
therefore

$$\begin{aligned} \frac{1}{B^2} \left(\text{Var} \sum_{i=1}^B T_i(n) \right) &= \frac{1}{B^2} \left[B \sigma^2 + B(B-1) \rho \sigma^2 \right] \\ &= \frac{1}{B^2} \left(B \sigma^2 (1 + (B-1)\rho) \right) \\ &= \frac{1}{B^2} B \sigma^2 (1 - \rho + B\rho) \\ &= \frac{1}{B} \sigma^2 (1 - \rho + B\rho) \\ &= \frac{(1 - \rho) \sigma^2}{B} + \rho \sigma^2 \end{aligned}$$

$$\therefore \boxed{\text{Var} \left(\frac{1}{B} \sum_{i=1}^B T_i(n) \right) = \rho \sigma^2 + \frac{(1 - \rho) \sigma^2}{B}} \text{ has been proved.}$$

3. a
Circle all the points whose weights will increase as a result of incorporating the first Stamp. Briefly Explain the Selection.

* As a result of incorporating the first decision Stamp, the decision boundary classified the data points as following:



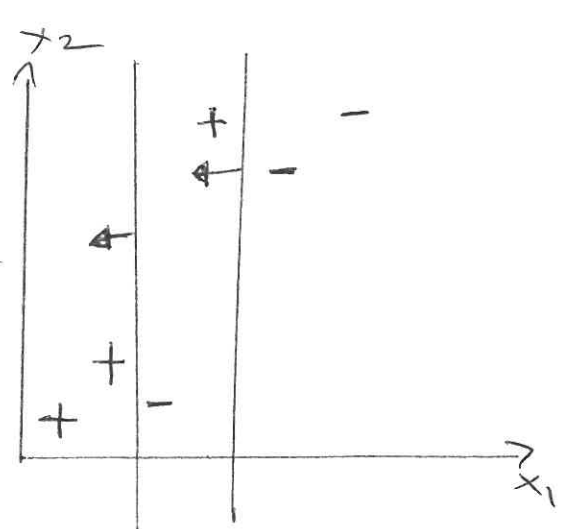
1. 3 positive data points Classified as positive (Correct Classification)
2. 2 negative data points Classified as negative (Incorrect Classification)
3. 1 negative point Classified as positive which is incorrect, highlighted in circle.

* In an attempt to make the future Stamp(s) to classify this misclassified data point properly, the first decision Stamp will assign higher weight to this data point.

As a result, the weight of misclassified point will increase thereby giving it higher importance in next Stamp towards a better Classification.

3.b Draw a Stump that we could select for next boosting iteration. Draw both decision boundary and positive orientation.

As first Stump gave higher weight to misclassified $\ominus$ (from 3.a), the Stump in second iteration aims to classify that $\ominus$ point into a negative side of the decision boundary.



This can be achieved by

- a) creating a horizontal split on x_2 such that $\ominus$ will be below the horizontal line with positive arrow pointing upwards.
- b) or by performing one more vertical split on x_1 such that $\ominus$ point in 3.a will be on the right side of the line. i.e. in the negative region of the new decision boundary. The arrow representing side of the decision boundary will be pointing towards left ($\leftarrow$) as shown in the figure above.

Since Vertical line (b) seem to misclassify less no. of points than the horizontal line described in a, I went with vertical line as the plausible solution.

3. c will $\hat{w}_2 > \hat{w}_1$, explain.

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* Stump 1 classified 1 out of 6 data points incorrectly.

$$\text{We know } \hat{w}_t = \frac{1}{2} \ln \left(\frac{1 - \text{weighted error}(t)}{\text{weighted error}(t)} \right)$$

Initially all data points have the weight of $1/n$ i.e. $1/6$.

$$\hat{w}_1 = \frac{1}{2} \ln \left(\frac{5/6}{1/6} \right) = \ln(\sqrt{5}) = \underline{\underline{0.1667}}$$

* At the end of 1 iteration, Stump 1 gave a higher weight of 0.5 to the misclassified point and lower weight of 0.1 to the remaining 5 rightly classified points.

* Stump 2 classified the previously misclassified point (the one with weight of 0.5) correctly but it possibly misclassified a different data point that was classified correctly by previous Stump and this point has a weight of 0.1.

$$\hat{w}_2 = \frac{1}{2} \ln \left(\frac{1 - 0.1}{0.1} \right) = \frac{1}{2} \ln \left(\frac{0.9}{0.1} \right) = \ln(\sqrt{9})$$

$$\text{Clearly } \ln(\sqrt{9}) > \ln(\sqrt{5})$$

Hence $\hat{w}_2 > \hat{w}_1$. This is due to the

advantage that Stump 2 has, for fixing the mistake that was classified with higher weight by previous Stump even though it made a different mistake its impact is low on $\hat{w}_2$.

4. a $F(x) = \text{Sign} \left\{ \sum_{t=1}^T \hat{w}_t \cdot f_t(x) \right\} = \text{Sign} \{ f(x) \}$ where $f(x) = \sum_{t=1}^T \hat{w}_t f_t(x)$ (16)

Show $\epsilon_{\text{training}} := \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{F(x^{(i)}) \neq y^{(i)}\}} \leq \frac{1}{n} \sum_{i=1}^n \exp(-f(x^{(i)}) \cdot y^{(i)})$

where $\mathbb{1}_{\{F(x^{(i)}) \neq y^{(i)}\}}$ is 1 if $F(x^{(i)}) \neq y^{(i)}$; otherwise 0.

Solution:-

We need to prove the following as part of solutioning.

① $\exp(-f(x^{(i)}) \cdot y^{(i)}) \geq 1$; if $F(x^{(i)}) \neq y^{(i)}$

② $\exp(-f(x^{(i)}) \cdot y^{(i)}) > 0$; if $F(x^{(i)}) = y^{(i)}$

Case ① if $F(x^{(i)}) \neq y^{(i)}$; i.e., the ensemble classified incorrectly

As $y^{(i)} \in (-1, 1)$; when $F(x^{(i)}) \neq y^{(i)}$, both $f(x^{(i)})$ and $y^{(i)}$ have opposite signs thereby making their product less than zero

$$\text{i.e. } f(x^{(i)}) \cdot y^{(i)} < 0$$

$$\Rightarrow -f(x^{(i)}) \cdot y^{(i)} > 0$$

$$\Rightarrow e^{-f(x^{(i)}) \cdot y^{(i)}} > e^0$$

$$\therefore e^{-f(x^{(i)}) \cdot y^{(i)}} > 1$$

i.e. $\boxed{1 < e^{-f(x^{(i)}) \cdot y^{(i)}}}$ has been proved.

Case 2 If $F(x^{(i)}) = y^{(i)}$ i.e. ensemble classified correctly

i.e. both $f(x^{(i)})$ and $y^{(i)}$ have same signs.

$$f(x^{(i)}) \cdot y^{(i)} > 0$$

$$-f(x^{(i)}) \cdot y^{(i)} < 0$$

$$\Rightarrow e^{-f(x^{(i)}) \cdot y^{(i)}} < e^0$$

$\Rightarrow \boxed{1 > e^{-f(x^{(i)}) \cdot y^{(i)}}}$ has been proved.

4.2

therefore we can say that, summing over all i & dividing by n ^{① ②}

$\varepsilon_{\text{training}}$

$$\frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{F(x^{(i)}) \neq y^{(i)}\}} \leq \frac{1}{n} \sum_{i=1}^n \exp(-f(x^{(i)}) \cdot y^{(i)})$$

has been proved by above ① and ② cases.

4.5)

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$$\alpha_{t+1}^{(i)} = \frac{\alpha_t^{(i)} \exp(-\hat{w}_t f_t(x^{(i)}) y^{(i)})}{Z_t}$$

$$Z_t = \sum_{i=1}^n \alpha_t^{(i)} \exp(-\hat{w}_t f_t(x^{(i)}) y^{(i)}); \text{ Initially } \alpha_1^{(i)} = 1/n$$

Show $\frac{1}{n} \sum_{i=1}^n \exp(-f(x^{(i)}) y^{(i)}) = \prod_{t=1}^T Z_t$.

Solution

$$\alpha_{t+1}^{(i)} = \frac{\alpha_t^{(i)} \exp(-\hat{w}_t f_t(x^{(i)}) y^{(i)})}{Z_t}$$

$$\alpha_t^{(i)} = \frac{\alpha_{t-1}^{(i)} \exp(-\hat{w}_{t-1} f_{t-1}(x^{(i)}) y^{(i)})}{Z_{t-1}}$$

.....

$$\alpha_2^{(i)} = \frac{\alpha_1^{(i)} \exp(-\hat{w}_1 f_1(x^{(i)}) y^{(i)})}{Z_1}$$

there fore

$$\alpha_{t+1}^{(i)} = \frac{\alpha_1^{(i)} \exp(-\hat{w}_1 f_1(x^{(i)}) y^{(i)})}{Z_1} \cdot \frac{\exp(-\hat{w}_2 f_2(x^{(i)}) y^{(i)})}{Z_2} \cdot \dots \cdot \frac{\exp(-\hat{w}_t f_t(x^{(i)}) y^{(i)})}{Z_t}$$

$$\alpha_{t+1}^{(i)} = \frac{1}{n} \cdot \frac{\exp\left(-\sum_{t=1}^T \hat{w}_t f_t(x^{(i)}) y^{(i)}\right)}{\prod_{t=1}^T Z_t}$$

As Reason :-
 $e^x \cdot e^y \cdot e^k = e^{x+y+k}$
 Also $\alpha_1 = 1/n$

4.b

$$\sum_{i=1}^m \alpha_{t+1}^{(i)} = 1 \quad \left[\text{as we are summing normalized weights} \right] \quad (19)$$

therefore by applying $\sum_{i=1}^m$ to $\alpha_{t+1}^{(i)}$

$$1 = \frac{1}{n} \cdot \sum_{i=1}^m \frac{\exp(-f(x^{(i)}) \cdot y^{(i)})}{\prod_{t=1}^T Z_t}$$

therefore

$$\prod_{t=1}^T Z_t = \frac{1}{n} \sum_{i=1}^m \exp(-f(x^{(i)}) \cdot y^{(i)})$$

has been proved.

4.C.i

(20)

$$\text{Let } \varepsilon_t = \sum_{i=1}^n \varphi_t^{(i)} \mathbb{1}_{\{t_t(x^{(i)}) \neq y^{(i)}\}}$$

$$z_t = (1 - \varepsilon_t) \exp(-\hat{w}_t) + \varepsilon_t \exp(\hat{w}_t)$$

find value of $\hat{w}_t$ that minimizes z_t ; optimal value

$$z_t^{\text{opt}} = \frac{2}{\sqrt{1 - \varepsilon_t}}$$

Solutioning :-

To find $\hat{w}_t$ that minimizes z_t ; let's find $\frac{dz_t}{d\hat{w}_t}$ & set to zero.

$$\frac{dz_t}{d\hat{w}_t} = \frac{d}{d\hat{w}_t} \left[(1 - \varepsilon_t) \exp(-\hat{w}_t) + \varepsilon_t \exp(\hat{w}_t) \right]$$

$$= -1 \cdot (1 - \varepsilon_t) \cdot \exp(-\hat{w}_t) + 1 \cdot \varepsilon_t \exp(\hat{w}_t)$$

$$\frac{dz_t}{d\hat{w}_t} = (\varepsilon_t - 1) \cdot \exp(-\hat{w}_t) + \varepsilon_t \exp(\hat{w}_t)$$

closed form solution $\frac{dz_t}{d\hat{w}_t} = 0$

$$(\varepsilon_t - 1) \cdot \exp(-\hat{w}_t) + \varepsilon_t \exp(\hat{w}_t) = 0$$

$$(1 - \varepsilon_t) \exp(-\hat{w}_t) = \varepsilon_t \exp(\hat{w}_t)$$

$$\frac{(1 - \varepsilon_t)}{\varepsilon_t} = \exp(2\hat{w}_t)$$

$$e^{2\hat{w}_t} = \frac{(1 - \varepsilon_t)}{\varepsilon_t}$$

$$\Rightarrow \hat{w}_t = \frac{1}{2} \ln \left(\frac{1 - \varepsilon_t}{\varepsilon_t} \right)$$

u.c.1

$$\therefore \hat{w}_t = \frac{1}{2} \ln \left(\frac{1 - \varepsilon_t}{\varepsilon_t} \right)$$

(21)

The above $\hat{w}_t$ value optimizes i.e. minimizes the value of Z_t . Hence by plugging in above $\hat{w}_t$ in Z_t we should be able to derive Z_t^{opt} .

Let's first derive the values of $e^{-\hat{w}_t}$ and $e^{\hat{w}_t}$.

$$e^{-\hat{w}_t} = e^{-\frac{1}{2} \ln \left(\frac{1 - \varepsilon_t}{\varepsilon_t} \right)} = \sqrt{\frac{\varepsilon_t}{1 - \varepsilon_t}}$$

$$e^{\hat{w}_t} = e^{\frac{1}{2} \ln \left(\frac{1 - \varepsilon_t}{\varepsilon_t} \right)} = \sqrt{\frac{1 - \varepsilon_t}{\varepsilon_t}}$$

Substituting these values in Z_t for getting Z_t^{opt}

$$Z_t^{\text{opt}} = (1 - \varepsilon_t) \cdot \sqrt{\frac{\varepsilon_t}{1 - \varepsilon_t}} + \varepsilon_t \sqrt{\frac{1 - \varepsilon_t}{\varepsilon_t}}$$

therefore

$$Z_t^{\text{opt}} = \sqrt{(1 - \varepsilon_t) \cdot \varepsilon_t} + \sqrt{\varepsilon_t \cdot (1 - \varepsilon_t)}$$

$$Z_t^{\text{opt}} = 2 \sqrt{\varepsilon_t (1 - \varepsilon_t)}$$

has been proved.

4.(ii) Assume we chose z_t as z_t^{opt} in part (i), then (22)
 rewrite $\varepsilon_t = \frac{1}{2} - \gamma_t$ where $\gamma_t > 0$ implies better than random and $\gamma_t < 0$ implies worse than random. Show that

$$z_t \leq \exp(-2\gamma_t^2)$$

we know $z_t^{\text{opt}} = 2\sqrt{\varepsilon_t(1-\varepsilon_t)}$

Choosing z_t as z_t^{opt} and substituting $\varepsilon_t = \frac{1}{2} - \gamma_t$

$$z_t = 2\sqrt{\left(\frac{1}{2} - \gamma_t\right)\left(1 - \left(\frac{1}{2} - \gamma_t\right)\right)}$$

$$= 2\sqrt{\left(\frac{1}{2} - \gamma_t\right)\left(\frac{1}{2} + \gamma_t\right)}$$

$$= 2\sqrt{\frac{1}{4} - \gamma_t^2}$$

$$z_t = \sqrt{1 - 4\gamma_t^2} \quad \text{--- (1)}$$

Applying $\ln$ function on both sides

$$\ln(z_t) = \frac{1}{2} \ln(1 - 4\gamma_t^2)$$

Given hint $\ln(1-x) \leq -x$;

Considering $-4\gamma_t^2$ as x .

$$\ln(1 - 4\gamma_t^2) \leq -4\gamma_t^2$$

multiplying with $\frac{1}{2}$ on both sides of inequality.

$$\frac{1}{2} \ln(1 - 4\gamma_t^2) \leq -2\gamma_t^2$$

$$\ln(\sqrt{1 - 4\gamma_t^2}) \leq -2\gamma_t^2$$

$$\ln(z_t) \leq -2\gamma_t^2 \quad \left\{ \begin{array}{l} \text{as we know from (1)} \\ z_t = \sqrt{1 - 4\gamma_t^2} \end{array} \right\}$$

$$\boxed{z_t \leq e^{-2\gamma_t^2}} \text{ has been proved.}$$

4.C.iii Show if classifier is better than random i.e $\gamma_t > \gamma$ (23)
 for all t and $\gamma > 0$, then $\epsilon_{\text{training}} \leq \exp(-2T\gamma^2)$

Solution :-

From 4.a and 4.b, we already proved that

$$\epsilon_{\text{training}} \leq \prod_{t=1}^T z_t \quad [\gamma_t > \gamma \text{ for all } t, \gamma > 0]$$

$$\epsilon_{\text{training}} \leq \prod_{t=1}^T \exp(-2\gamma_t^2) \quad [\text{from 4.c.ii}]$$

$$\epsilon_{\text{training}} \leq \exp\left(\sum_{t=1}^T -2\gamma_t^2\right)$$

$$\boxed{\epsilon_{\text{training}} \leq \exp\left(\sum_{t=1}^T -2\gamma_t^2\right)} \quad \text{--- (1)} \quad (\text{coz } e^a \cdot e^b = e^{a+b})$$

Let's analyze internal expression within exp function

$$\sum_{t=1}^T -2\gamma_t^2 = -2[\gamma_1^2 + \gamma_2^2 + \dots + \gamma_t^2]$$

Since we know $\gamma_1 > \gamma$, $\gamma_2 > \gamma$, ..., $\gamma_t > \gamma$.

$$\sum_{t=1}^T 2\gamma_t^2 > T \cdot \gamma^2$$

$$\text{ie } \sum_{t=1}^T -2\gamma_t^2 < -2T\gamma^2$$

$$\text{therefore } \boxed{e^{\sum_{t=1}^T -2\gamma_t^2} < e^{-2T\gamma^2}} \quad \text{--- (2)}$$

putting (1) & (2) side by side

$$\epsilon_{\text{training}} \leq \exp\left(\sum_{t=1}^T -2\gamma_t^2\right) < \exp(-2T\gamma^2)$$

Hence $\boxed{\epsilon_{\text{training}} \leq \exp(-2T\gamma^2)}$ is proved